

Contact

tschehd@gmail.com

www.linkedin.com/in/dennis-treder-tschechlov (LinkedIn)

Top Skills

IBM Cloud

DevOps

Infrastructure as code (IaC)

Certifications

IBM Machine Learning Specialist - Advanced

HashiCorp Certified: Terraform Associate (003)

How to Win a Data Science Competition: Learn from Top Kagglers

Generative AI with Large Language Models

Introduction to Deep Learning

Publications

Exploiting domain knowledge to address class imbalance and a heterogeneous feature space in multi-class classification

AutoML4Clust: Efficient AutoML for Clustering Analyses

Efficient Exploratory Clustering Analyses with Qualitative Approximations

SDRank: A Deep Learning Approach for Similarity Ranking of Data Sources to Support User-Centric Data Analysis

ML2DAC: Meta-Learning to Democratize AutoML for Clustering Analysis

Dennis Treder-Tschechlov, PhD

Data&AI || Platform Engineer || Client Engineering
Sindelfingen, Baden-Württemberg, Germany

Summary

I am a research scientist at the University of Stuttgart. I am fascinated by novel methods to automate and democratize machine learning approaches. In my PhD thesis, I am focusing on automating the unsupervised task of clustering analyses for novice analysts. This has led to publications at international top-conferences and journals such as SIGMOD or VLDB-journal. I love to learn new trending topics regarding data management, machine learning & AI. For instance, I have completed Coursera Courses regarding Deep Learning and Reinforcement Learning.

Experience

IBM

Platform Engineer || Data&AI and Automation || Client Engineering
January 2025 - Present (8 months)
Böblingen, Baden-Württemberg, Germany

Institut für Parallele und Verteilte Systeme

Research Scientist for Data & AI
March 2020 - February 2025 (5 years)
Stuttgart

- Research on new methods for democratizing clustering analyses with AutoML, Meta-Learning, and Ensemble methods
- Published 4 first-author papers, two of them at the at top-tier conferences SIGMOD and VLDB
- 8 published co-author publications, two of them at the top-tier journal VLDB-Journal
- Organization of events with industry partners Bosch, IBM, and Zeiss
- Guided hands-on assignments for students about machine learning concepts and NoSQL systems
- Supervision of over 20 student works, including student theses, practical courses, data science projects, and seminar works

Software Campus

Lead of Software Campus Project „VALID-PARTITION“

January 2022 - June 2024 (2 years 6 months)

Stuttgart, Baden-Württemberg, Deutschland

- Research grant of 100.000€ from German Federal Ministry of Education and Research (BMBF)
- Cooperation with industry partner Trumpf SE + Co. KG
- Managed budget, resources, and progress of the project
- Application of developed methods for a real-world use case
- Supervision of research scientists and student assistants

Robert Bosch Smart Home GmbH

Working Student Automated Testing

April 2019 - January 2020 (10 months)

Stuttgart, Baden-Württemberg, Deutschland

- Implementation of tools for process and workflow optimization
- Automated and manual testing of Android and iOS app

Universität Stuttgart

Student Research Assistant

December 2017 - March 2019 (1 year 4 months)

Stuttgart

- Research on state-of-the-art parameter optimization algorithms for clustering analyses
- Implementation of selected state-of-the-art algorithms for clustering on Apache Spark

Novatec Software

Working Student Software Development

February 2017 - February 2018 (1 year 1 month)

Stuttgart

Development, testing, and code reviews of web application for human resource management.

Universität Stuttgart

Tutor „Theoretical Foundations of Computer Science“

August 2015 - March 2017 (1 year 8 months)

Stuttgart

Explanations of lecture contents and additional exercises

Education

Universität Stuttgart

Ph.D., Data Science/ Machine Learning · (March 2020 - October 2024)

Universität Stuttgart

Master of Science - MS, Software Engineering · (October 2017 - December 2019)

Universität Stuttgart

Bachelor of Science - BS, Software Engineering · (October 2013 - October 2017)